# DIABETES IN LABOR: INSULIN AND IV FLUIDS

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Companion to Diabetes in pregnancy · CSM OB insulin infusion protocol (CHRISTUS Health) · ACOG PB 190/201 · ADA Standards 2026 §6, §15–16 · Labor and delivery · Decision support only, not unit policy

**HOW TO READ THIS CHART:** Step 1 on admission. Each FSBG: find the row, set the insulin rate AND the fluid. **Below 70 = low: rescue box.** Hexagons are questions; shaded rectangles are actions.

**SHADING KEY:** WHITE = monitor / set-up · LIGHT GRAY = admission, no-drip rule · MID GRAY = insulin running · DARK GRAY = hypoglycemia, DKA

### STEP 0: DIABETES IN LABOR

Diabetes (GDM A1/A2, type 2, type 1) admitted for labor, induction or scheduled cesarean. Unsure of type AND on home insulin, insulin pump, or prior DKA: treat as type 1 for basal (never a basal gap) and get senior/endocrine review now. Not on any insulin and no DKA history: diet/oral rule (no drip).

### STEP 1: ON ADMISSION (CSM) · LIGHT GRAY

Stop all insulin, including SC insulin pumps (unless a signed CSM exception), and oral agents. NPO except 8 oz ice chips every 4 hours. POC glucose; BMP and urine ketones if FSBG >160. Type 1: start the IV floor before stopping SC/pump, or keep the glargine/detemir already given; never a gap. Night-before/morning home doses and steroid adjustment: per the CSM provider/pharmacy order set (this app gives no home doses).

### DRIP SET-UP · WHITE

Regular insulin (Myxredlin) 100 units/100 mL, so units/h = mL/h; range 0.5–5 units/h. PE-lined tubing: fill with insulin, wait 20 minutes, then flush. Smart pump, dedicated line, independent double-check. High-alert medication. The CSM protocol runs on a provider order.

### MONITORING · WHITE

FSBG before starting IV fluids.
Insulin drip running: FSBG every 1 hour, regardless of dilation (Michael override of CSM q2h latent; ACOG PB 201).
Insulin-requiring, drip not yet running: every 2 hours in latent labor or NPO before delivery; every 1 hour at 4–6 cm or with regular contractions (CSM).
Diet-controlled GDM or oral agents only (no drip): every 4 hours until active labor, then every 2 hours; recheck 1 hour after any sliding-scale dose (Michael decision 11 Nov 2026). Active labor = 6 cm or more (ACOG).
After any hypoglycemia rescue: recheck in 15 minutes (go-live resolution 2).
Hospital point-of-care glucose for insulin decisions; CGM for trends/alarms only.

### TYPE 1: NEVER ZERO BASAL · MID GRAY

At 70–90 the floor is 0.5 units/h with D5LR 125 mL/h and provider notification; the provider may set 0.5–1 units/h by trend (prevents ketosis; not glucose correction). Clinician override of the CSM chart (CSM: hold), approved by Dr. Carozza 11 Nov 2026. 61–69: keep the floor AND give the hypoglycemia rescue; recheck in 15 min. The IV floor is not needed only if effective long-acting SC basal (glargine/detemir given within its duration) is on board, or a pump is continued under a signed CSM exception. Otherwise the pump stops only after the IV floor is running. Do not stack. Type 1 insulin stops only for FSBG ≤60 or symptoms, as an emergency pause: dextrose continues, rescue, recheck in 15 minutes, resume 0.5 units/h once >70 and rising (rising = higher than the previous check); a pause of about 1 hour or longer needs provider review.

### STEP 2: FSBG NOW — WHICH ROW? (CSM NOMOGRAM, MAINTENANCE 125 mL/h)

CSM nomogram; maintenance fluid 125 mL/h. FSBG is read as a whole number in mg/dL. If a value has a decimal, round DOWN (60.5 → 60). 60 is the ≤60 row; 90 holds; 91 starts 0.5 units/h; 120 stays on D5LR; 121 switches to LR. Meter "LO": rescue now. Meter "HI": call provider now, DKA screen, lab glucose (drip group: >250 row; no-drip group: no drip unless ordered). No bolus. Maximum 5 units/h. Never use this chart to treat DKA.

**BELOW 70:** Any FSBG below 70, or symptoms at any value: hypoglycemia rescue now (rescue box). 61–90 means "no insulin increase," not "no action."

| FSBG (mg/dL) | GDM A2 / type 2 on insulin | Type 1: insulin | Fluid at 125 mL/h |
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| **≤60 (CSM "<60" row; 60 included)** | No insulin. Contact provider. Hypoglycemia rescue now; recheck 15 min. | Emergency pause (dextrose continues); resume 0.5 units/h once >70 and rising (rising = higher than the previous check). | **D5LR** |
| **61–90 (insulin hold band; 61–69 = LOW, rescue)** | Hold insulin (no added insulin). 61–69: hypoglycemia rescue now, recheck in 15 min, notify provider. | 0.5 units/h floor (provider may set 0.5–1) + notify provider. 61–69: keep the floor AND give the hypoglycemia rescue; recheck in 15 min. Clinician override of the CSM chart (CSM: hold). | **D5LR** |
| **91–120** | **0.5 units/h** | 0.5 units/h | **D5LR** |
| **121–150** | **1 units/h** | 1 units/h; ketones at LR start, then q4h (Doc proposal) | **LR** |
| **151–180** | **2 units/h** | 2 units/h; ketones at LR start, then q4h (Doc proposal) | **LR** |
| **181–220** | **3 units/h** | 3 units/h; ketones at LR start, then q4h (Doc proposal) | **LR** |
| **221–250** | **4 units/h** | 4 units/h; ketones at LR start, then q4h (Doc proposal) | **LR** |
| **>250** | **5 units/h (maximum; notify provider)** | 5 units/h; ketones at LR start, then q4h (Doc proposal) | **LR** |

### DIET-CONTROLLED GDM (A1) OR ORAL AGENTS ONLY: NO DRIP · LIGHT GRAY

Diet-controlled GDM (A1) and anyone on oral agents ONLY (no home insulin): NO insulin drip. Any home insulin (with or without metformin/glyburide): drip group. Sliding scale insulin as needed, per provider order (no CSM scale in this app; no doses shown). FSBG every 4 h until active labor, then every 2 h; recheck 1 h after any sliding-scale dose. Active labor = 6 cm or more (ACOG). Call provider now for any single FSBG >180, any ketones, or vomiting/illness. The provider decides next steps: DKA screen (beta-hydroxybutyrate, VBG, BMP) and the CSM insulin drip only by provider order. If the drip is ordered, move to the drip group (hourly FSBG, CSM nomogram). Do not start the drip from this chart without an order. Low (<70): rescue box. Glyburide within the last 24 h: FSBG every 2 h until 24 h after the last dose. After any sliding-scale dose: recheck FSBG in 1 h, then every 2 h until 2 values in a row are 70–180 (or per provider), then back to the routine schedule. The drip is for insulin-requiring patients: A2 on insulin, type 2 on insulin, type 1 (Michael decision 11 Nov 2026).

### TYPE 1 ON PLAIN LR (≥121): KETONES · MID GRAY

Type 1 on plain LR: urine ketones (or beta-hydroxybutyrate) at the start of LR and every 4 h on LR (q4h = Doc proposal, not CSM), and now for nausea, vomiting, abdominal pain, fast breathing or 5 units/h with rising values. Any positive ketones: call provider; go to the DKA box; no ad hoc dextrose.

### HYPOGLYCEMIA <70 OR SYMPTOMS: RESCUE NOW (CSM ADULT HYPOGLYCEMIC PROTOCOL) · DARK GRAY

**D50 50 mL IV:** FSBG <50 with IV access: alert and NPO or unable to swallow (NPO = unable to swallow), OR questionable consciousness (CSM; pharmacy).
**D50 25 mL IV push:** FSBG 50–69 with IV access: alert and NPO or unable to swallow, OR questionable consciousness (CSM; pharmacy).
**Glutose 40% gel 30 g / 15 g buccal:** ALERT, no IV, can swallow safely: 30 g if <50; 15 g if 50–59 (CSM). 15 g also at 60–69 (Michael-approved addition; pending CSM pharmacy build).
**Glucagon (GlucaGen) 1 mg IM, ONCE:** Only if ALERT AND no IV AND cannot swallow, FSBG <70; OR questionable consciousness, no IV, FSBG <60 (CSM). Also questionable consciousness, no IV, 60–69 (Michael-approved addition; pending CSM pharmacy build).
**No IV in any rescue:** Get IV access now (urgent provider / rapid-response call if consciousness is impaired).
Recheck FSBG in 15 minutes. Still <70: repeat D50 or gel per the matching row. Glucagon is ONCE only: if still <70 15 minutes after glucagon, get IV access for D50 and call the provider / rapid response. Notify the provider after every rescue.
Below 54 or any symptoms: rescue and call the provider immediately.
Symptoms but FSBG ≥70: repeat POC, send lab glucose, call provider. If you cannot check, treat as low.
D10 is not in the CSM protocol: not used here. D50 is a vesicant: patent vein, then flush. D5LR continues during rescue.
GDM/type 2: hold IV insulin during rescue and restart per the nomogram row. Type 1: resume 0.5 units/h once >70 and rising (rising = higher than the previous check).

### SICK, KETONES, VOMITING? → DKA · DARK GRAY

Pregnancy DKA can occur below 200 mg/dL; there is no glucose gate for ketone testing when ill. Check beta-hydroxybutyrate, VBG, electrolytes/K+, creatinine and anion gap; maternal assessment and fetal monitoring. Use the CSM DKA order set, not this chart. Reference only, per CSM DKA order set: insulin 0.1 units/kg/h once K+ is safe. Type 1 on plain LR with positive ketones comes here; no ad hoc dextrose. Delivery is not DKA treatment.

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| **A · AFTER DELIVERY (CSM)** | Notify the provider when delivery is complete to start the CSM Postpartum Diabetes Management Order Set (doses there). GDM: stop insulin at placenta; 75-g 2-hour OGTT at 4–12 weeks postpartum. Type 1: basal always covered. Continue IV insulin 1–2 hours after the first SC basal dose; then stop D5LR and insulin together. |
| **B · NEWBORN** | Tell the nursery the diabetes type, treatment, last maternal glucose and steroids. Stable baby: feed within 1 hour of birth; the nursery team owns the neonatal glucose protocol. |
| **C · STEROIDS / TIMING** | Steroids raise glucose up to 5 days: more FSBG checks; follow the nomogram. SMFM #58: no late-preterm (34+0–36+6) steroids in pregestational diabetes. Timing and EFW ≥4,500 g counseling: Diabetes in pregnancy card. |
| **D · NOTIFY** | RN to OBHG hospitalist: any FSBG <70 or any rescue; 61–90 on a type 1 floor; 5 units/h with rising values; ketones; any gap in type 1 basal; diet/oral group: >180, ketones or illness. |

**Doses to be checked against CSM pharmacy build.** Decision support only, not orders or unit policy. FSBG = fingerstick blood glucose; BHB = beta-hydroxybutyrate; POC = point of care. **Cite:** CHRISTUS Health OB insulin infusion protocol (CSM, transcribed 11 Nov 2026); ACOG PB 190 and 201; ADA Standards of Care 2026 §6, §15–16; SMFM Consult #58.

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